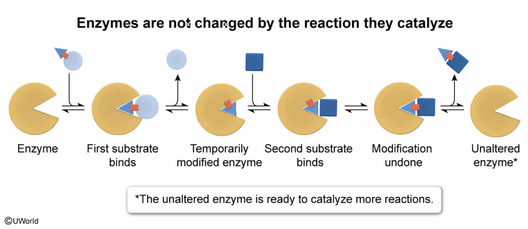
All of the following are true of enzyme-catalyzed reactions EXCEPT:

- A. the presence or absence of a post-translational modification may alter the activation energy E_a of the reaction. [22%]
- B. the enzyme does not affect the net reaction rate if the ratio of products to reactants equals K_{eq} for the reaction. [34%]
- ✔ C. substrates can covalently modify the enzyme to cause a permanent decrease in the enzyme's turnover number k_{cat}. [28%]
- D. the saturation of enzyme active sites by substrate molecules limits the maximum reaction velocity V_{max}. [14%]

Correct
29% answered correctly

Explanation:



Enzymes are biological catalysts that speed up chemical reactions, allowing them to occur under the mild physiological conditions present within living organisms. Enzymes do this by **stabilizing** the transition state of the reaction, which lowers the reaction's **activation energy** $\boldsymbol{E_a}$; however, enzymes do not change the **equilibrium** of the reaction. If a system is already at equilibrium (ie, if $\frac{[\text{Products}]}{[\text{Reactants}]} = K_{\text{eq}}$), the addition of an enzyme will **not** affect the net rate of the reaction, and the net rate will remain at zero **(Choice B)**.

If a system is *not* at equilibrium, the reaction rate can still be limited through other means. The number of enzymes in a system is finite, and therefore the system contains a limited number of active sites. Once all active sites are bound to substrate, the reaction rate cannot increase unless more enzyme is added. This condition is called enzyme saturation, and the maximal reaction rate (ie, maximal reaction velocity) is $\boldsymbol{V_{max}}$ **(Choice D)**.

Living systems can also **regulate** enzymatic reaction rates through posttranslational modifications. For example, phosphorylation (or dephosphorylation) is a common way to **activate or inactivate** different enzymes. Active enzymes act as catalysts by lowering the E_a of a reaction; inactive enzymes do *not* act as catalysts, and therefore do not lower the E_a. Therefore, post-translational modifications may alter the E_a of the reaction **(Choice A)**.

The question asks which option is **not** true of enzyme-catalyzed reactions. A crucial property of enzymes (and of catalysts in general) is that they are **not permanently altered** by the chemical reaction they catalyze. Although an enzyme may experience temporary changes during its catalytic cycle (eg, covalent catalysis, acid–base catalysis, oxidation–reduction catalysis), these changes are **reversed** or **undone** by the end of the reaction. Any permanent change to the enzyme would imply that the *enzyme* is another reactant rather than a catalyst. Note that suicide inhibitors (also called mechanism-based inhibitors) are molecules that *resemble* substrates and covalently modify enzymes, but these inhibitors are *not* substrates themselves.

Because **no permanent change is expected** when an enzyme acts on its substrate, the turnover number k_{cat} is **not expected to permanently change**.

Educational objective:
Enzymes catalyze reactions in living systems by lowering the activation energy E_a. Enzyme rates are affected by the regulation of available reactant and product concentrations, by regulating enzyme concentration, or by regulating post-translational modifications. Enzymes may be covalently modified temporarily during a reaction, but they are not permanently affected by the reaction they catalyze.

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